

PEIZHONG JU

PARTICULARS

Current Position

Assistant Professor

Department of Computer Science at **University of Kentucky**

Lexington, KY, USA

2024 - present

Previous Positions

Postdoctoral Scholar

ECE department at **The Ohio State University**

NSF AI Institute on Future Edge Networks and Distributed Intelligence (**AI-EDGE**)

Columbus, OH, USA

2021 - 2024

Education

Purdue University

Ph. D. in Electrical and Computer Engineering

West Lafayette, IN, USA

2016 - 2021

Peking University

B. Sc. in Electrical Engineering

Beijing, China

2012 - 2016

Research Interests

My research interests span the areas of machine learning, smart grid, and wireless communication. My current research topics are about analyzing the performance of various machine learning models (such as generative AI) and designing new/better machine learning algorithms.

ACADEMIC HONORS

- Best Paper Award, 13th ACM International Conference on Future Energy Systems (ACM e-Energy 2022)
- The Bilsland Dissertation Fellowship, Purdue University, 2021
- Best Paper Award Finalist of 9th ACM International Conference on Future Energy Systems (ACM e-Energy 2018)
- Outstanding Graduate Award of EE department of Peking University, 2016 (the only one that year)
- Outstanding Graduate Award of Peking University, 2016
- IEEE Student Grant, 2015
- Research Excellence Award, Peking University, 2015
- China National Scholarship, 2015
- Samsung Scholarship, 2014
- EMC Scholarship, 2013

RESEARCH EXPERIENCE

- **Postdoctoral Scholar, the Ohio State University**, supervisor: Prof. Ness B. Shroff.
 - Multi-agent Reinforcement Learning, Multi-task supervised learning, Generalization Performance Analysis. (Oct. 2021 - Aug. 2024)
- **Research Assistant, Purdue University**, advisor: Prof. Xiaojun Lin.
 - Overfitting in Machine Learning (Oct. 2019 - Sept. 2021)

- Market Design for Renewable in Power Distribution Level (Aug. 2018 - Oct. 2019)
- Voltage Control in Power Grid with Renewable (Aug. 2016 - Aug. 2018)
- **Undergraduate Research Assistant, Peking University, China**, advisor: Prof. Xiang Cheng.
 - Antenna Grouping for Massive MIMO (Sep. 2015 - June 2016)
 - Self-Interference Cancellation for Full-Duplex (Oct. 2013 - Sep. 2015)

PUBLICATIONS

Conference Papers

1. **Peizhong Ju**, Haibo Yang, Jia Liu, Yingbin Liang, and Ness B. Shroff, “Can We Theoretically Quantify the Impacts of Local Updates on the Generalization Performance of Federated Learning?” *25th International Symposium on Theory, Algorithmic Foundations, and Protocol Design for Mobile Networks and Mobile Computing (MobiHoc’24)*.
2. Chengzhang Li, **Peizhong Ju**, Atilla Eryilmaz, and Ness B. Shroff, “Efficient Multi-dimensional Compression for Network-edge Classification,” *25th International Symposium on Theory, Algorithmic Foundations, and Protocol Design for Mobile Networks and Mobile Computing (MobiHoc’24)*.
3. **Peizhong Ju**, Arnob Ghosh, and Ness B. Shroff, “Achieving Fairness in Multi-Agent MDP Using Reinforcement Learning,” *12th International Conference on Learning Representations (ICLR’24)*.
4. Yining Li, **Peizhong Ju**, and Ness B. Shroff, “Achieving Sample and Computational Efficient Reinforcement Learning by Action Space Reduction via Grouping,” *12th International Conference on Learning Representations (ICLR’24)*.
5. Sen Lin, **Peizhong Ju (co-first author)**, Yingbin Liang, and Ness B. Shroff, “Theory on Forgetting and Generalization of Continual Learning,” *40th International Conference on Machine Learning (ICML’23)*.
6. **Peizhong Ju**, Yingbin Liang, and Ness B. Shroff, “Theoretical Characterization of the Generalization Performance of Overfitted Meta-Learning,” *11th International Conference on Learning Representations (ICLR’23)*.
7. **Peizhong Ju**, Xiaojun Lin, and Ness B. Shroff, “On the Generalization Power of Overfitted Three-Layer Neural Tangent Kernel Model,” *36th Conference on Neural Information Processing Systems (NeurIPS’22)*.
8. **Peizhong Ju**, Xiaojun Lin, and Jianwei Huang, “Distribution-Level Markets under High Renewable Energy Penetration,” *13th ACM International Conference on Future Energy Systems (e-Energy’22) (Best Paper Award)*.
9. **Peizhong Ju**, Xiaojun Lin, and Ness B. Shroff, “On the Generalization Power of Overfitted Two-Layer Neural Tangent Kernel Models,” *38th International Conference on Machine Learning (ICML’21) (Spotlight Presentation)*.
10. **Peizhong Ju**, Xiaojun Lin, and Jia Liu, “Overfitting Can Be Harmless for Basis Pursuit, But Only to a Degree,” *34th Conference on Neural Information Processing Systems (NeurIPS’20) (Spotlight Presentation)* (acceptance rate: 20%, spotlight presentations: 3%).
11. **Peizhong Ju**, and Xiaojun Lin, “Adversarial Attacks to Distributed Voltage Control in Power Distribution Networks with DERs,” *Proceedings of the 9th International Conference on Future Energy Systems (e-Energy’18) (Best Paper Award Finalist)*.
12. **Peizhong Ju**, Meng Zhang, Xiang Cheng, and Liuqing Yang, “Generalized spatial modulation with transmit antenna grouping for massive MIMO,” *Proceedings of 2017 IEEE International Conference on Communications (ICC’17)*, Paris, France, May 21-25, 2017.
13. **Peizhong Ju**, Meng Zhang, Xiang Cheng, Cheng-Xiang Wang, and Liuqing Yang, “Generalized spatial modulation with transmit antenna grouping for correlated channels,” *Proceedings of 2016 IEEE International Conference on Communications (ICC’16)*, Kuala Lumpur, Malaysia, May 22-27, 2016.
14. **Peizhong Ju**, Miaowen Wen, Xiang Cheng, and Liuqing Yang, “An Effective Self-Interference Cancellation Scheme for Spatial Modulated Full Duplex Systems,” *Proceedings of 2015 IEEE International Conference on Communications (ICC’15)*, London, UK, June 8-12, 2015.

Journal Papers

15. **Peizhong Ju**, Chengzhang Li (co-first author), Yingbin Liang, and Ness B. Shroff (the last two authors represent the entire AI-EDGE faculty team), “AI-EDGE: An NSF AI Institute for Future Edge Networks and Distributed Intelligence.” to appear in an upcoming issue of *AI Magazine*.
16. **Peizhong Ju**, Miaowen Wen, Xiang Cheng, and Liuqing Yang, “Achievable-Rate-Enhancing Self-Interference Cancellation for Full-Duplex Communications,” *IEEE Transactions on Wireless Communications*, vol. 17 no. 12, pp. 8473-8484, 2018.
17. Weilin Qu, Meng Zhang, Xiang Cheng, and **Peizhong Ju**, “Generalized Spatial Modulation With Transmit Antenna Grouping for Massive MIMO,” *IEEE Access*, vol. 5, pp. 26798-26807, 2017.

Pending Papers

18. Yuchen Liang, **Peizhong Ju**, Yingbin Liang, and Ness B. Shroff, “Non-asymptotic Convergence of Discrete-time Diffusion Models: New Approach and Improved Rate,” *arXiv preprint arXiv:2402.13901* (2024).
19. **Peizhong Ju**, Sen Lin, Mark S. Squillante, Yingbin Liang, and Ness B. Shroff, “Generalization Performance of Transfer Learning: Overparameterized and Underparameterized Regimes,” *arXiv preprint arXiv:2306.04901* (2023).
20. Chengzhang Li, **Peizhong Ju**, Atilla Eryilmaz, and Ness B. Shroff, “Two Levels Are All You Need: Simplifying Data Compression for Timely Edge Classification.”

Book Chapters

1. **Peizhong Ju**, Xiaojun Lin, and Ness B. Shroff, “On the Generalization Power of Overfitted Two-Layer Neural Tangent Kernel Models,” Chapter 3 of *Artificial Intelligence for Edge Computing*, Springer Nature, DOI: [10.1007/978-3-031-40787-1_3](https://doi.org/10.1007/978-3-031-40787-1_3), August 2023.

Patents

1. Xiang Cheng, **Peizhong Ju**, and Miaowen Wen, “A method of self-interference cancellation scheme for full duplex systems,” 2015100550190, State Intellectual Property Office of the People’s Republic of China.

INVITED TALKS

1. “Overview of Reinforcement Learning”, and “RL at the Edge: Fairness and Action Space Reduction” FutureG Summer Research Camp, Arizona State University, May 7, 2024.
2. “Achieving Fairness in Multi-Agent MDP Using Reinforcement Learning” AI-EDGE SPARKS Seminar, The Ohio State University, March 22, 2024.
3. “On the Generalization Power of the Overfitted Three-Layer Neural Tangent Kernel Model” The 15th International Conference of the ERCIM WG on Computational and Methodological Statistics (CM-Statistics 2022), King’s College London, 17-19 December 2022.
4. “Distribution Level Markets under High Renewable Energy Penetration” AIRS in the AIR Academic Series, Shenzhen Institute of Artificial Intelligence and Robotics for Society (AIRS), Shenzhen, China, August 23, 2022.
5. “On the Generalization Power of Overfitted Two-Layer Neural Tangent Kernel Models” Long Feng Science Forum, CUHK-Shenzhen, August 19, 2022.
6. “On the Generalization Power of Overfitted Two-Layer Neural Tangent Kernel Models” Joint Workshop between AI-EDGE and IBM, the Ohio State University, Columbus, June 15, 2022.
7. “On the Generalization Power of Overfitted Two-Layer Neural Tangent Kernel Models” TDAI deep learning summer school, the Ohio State University, Columbus, June 3, 2022.

MEDIA COVERAGE

- **The Ohio State University News** (Jul 20, 2023), [“Future AI algorithms have potential to learn like humans, say researchers.”](#) A research by Sen Lin, Peizhong Ju, Yingbin Liang, and Ness Shroff on “continual learning” will be presented at the 40th International Conference on Machine Learning, Honolulu, Hawaii.

TEACHING & MENTORING EXPERIENCE

- **Instructor.** CS 660: Topics in AI: Generative Machine Learning. Fall 2024, University of Kentucky.
- **Mentoring** several undergraduate students and one high school student in the summer research program held by AI-EDGE, 2024.
- **Secondary Instructor.** CSE-ECE 6461/6101: Introduction to Computer Communication Networks, Prof. Ness B. Shroff, Fall 2023, the Ohio State University.
- **Mentoring** a PhD student under the supervision of Prof. Ness B. Shroff and Prof. Atilla Eryilmaz at the Ohio State University (Dec. 2022 - present).
- **Mentoring** a female PhD student under the supervision of Prof. Ness B. Shroff at the Ohio State University (Oct. 2021 - present).
- **Mentoring** an undergraduate student and two high school students in the summer research program held by AI-EDGE, 2023.
- **Mentoring** an undergraduate student in Minority Serving Institutions (MSI) summer internships for AI-EDGE, 2022.
- **Teaching Assistant.** ECE 368: Data Structures, Prof. Cheng-Kok Koh, Spring 2020, Purdue University.
- **Teaching Assistant.** ECE 547: Introduction to Computer Communication Networks, Prof. Xiaojun Lin, Fall 2018 and Fall 2020, Purdue University.
- **Undergraduate Teaching Assistant.** Introduction to Computation, Prof. Yafei Dai, Fall 2012, Peking University, China.

SERVICE

- **Reviewer (Journals)**
 - IEEE Transactions on Information Theory (TIT)
 - IEEE Transactions on Signal Processing (TSP)
 - Transactions on Machine Learning Research (TMLR)
 - IEEE Transactions on Vehicular Technology (TVT)
 - IEEE Transactions on Wireless Communications (TWC)
 - IEEE Journal of Selected Topics in Signal Processing (J-STSP)
 - IEEE Transactions on Sustainable Computing (TSUSC)
 - IEEE Access
 - IEEE/ACM Transactions on Networking (TON)
 - Networks and Spatial Economics (NETS)
 - Internet Technology Letters
 - Transactions on Mobile Computing (TMC)
 - Frontiers of Information Technology & Electronic Engineering
- **Reviewer (Conferences)**
 - 38th Conference on Neural Information Processing Systems (NeurIPS'24)
 - 41st International Conference on Machine Learning (ICML'24)
 - 12th International Conference on Learning Representations (ICLR'24)
 - 37th Conference on Neural Information Processing Systems (NeurIPS'23)
 - 40th International Conference on Machine Learning (ICML'23)
 - 11th International Conference on Learning Representations (ICLR'23)
 - 36th Conference on Neural Information Processing Systems (NeurIPS'22)
 - 39th International Conference on Machine Learning (ICML'22)
 - 10th International Conference on Learning Representations (ICLR'22)
 - 35th Conference on Neural Information Processing Systems (NeurIPS'21)
 - 2021 National Conference on Communications (NCC'21)

- 38th International Conference on Machine Learning (ICML'21)
- IEEE Vehicular Technology Conference (VTC2015-Spring)
- IEEE International Conference on Communications (ICC'16)

- **Program Committee**

- 39th Annual AAAI Conference on Artificial Intelligence (AAAI'25)

LANGUAGES

Proficient in English and Mandarin Chinese.